VoltGuard weekly summary · region CEN

Report VG-WK-CEN-2026-W44 · generated 28 Oct 2026, 11:05 MYT · for revenue-protection managers, region CEN

Meter data: 24–30 Sep 2026
Field work: 21–28 Oct 2026

Simulated data (R0): not TNB data. Figures show how the system works, not real network performance.

This week in one paragraph. Unexplained energy in region CEN averaged 285 kWh a day, 2.5% lower than the week before, so the leak is steady, not growing. 2 of 3 zones are flagged; TX-CEN-002 alone accounts for 55% of the unexplained energy and draws a flat 24/7 load typical of mining rigs. 5 case(s) were opened and 0 closed this week; 2 inspection(s) were recorded.

- **Action 1:** Manager: add random routine checks: 2 so far is too few to measure the model's real hit rate (aim for at least 10).
- **Action 2:** Data team: confirm the TNB tariff so value at risk can be shown in RM.

1 · Key figures

2 of 3
Flagged zones
Unexplained energy above the flag rule for 7 days in a row
no change vs last week

285
Unexplained kWh per day
Average over the meter week, all zones with a transformer meter
−2.5% vs 292 the week before

~8,554
kWh a month at this level
285 × 30 days. Value in RM: [TNB tariff per kWh: to be supplied]
estimate, see section 3

5 / 0
Cases opened / closed
During the field-work week
10 closed in total

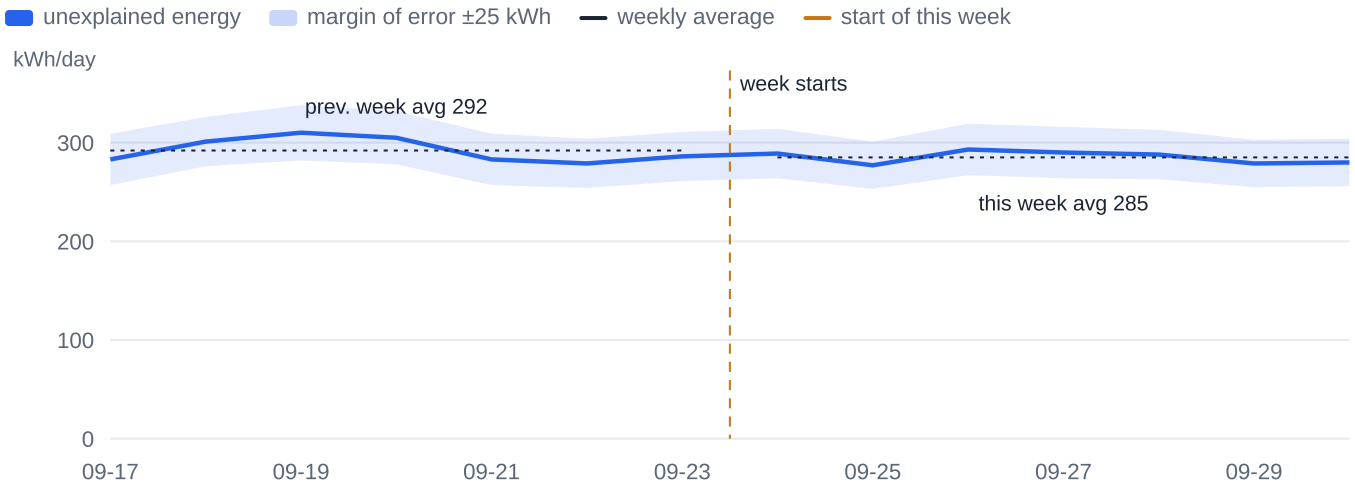
0
Overdue cases
Assigned and not visited within 72 h

0 · 0
Reports exported · open alerts
Reports after both approvals · unacknowledged data or model alerts
0 new alert(s) this week

Why two dates? Meter data reach VoltGuard daily, so energy figures cover the latest full week of readings (24–30 Sep 2026). Cases and inspections are counted over the last 7 calendar days (21–28 Oct 2026). In this prototype the simulated meter data end on 30 Sep 2026, which is why the two windows are 28 days apart; with live TNB data they would be one or two days apart.

2 · Is energy still going missing?

Unexplained energy per day, whole region, last 14 days (kWh/day)

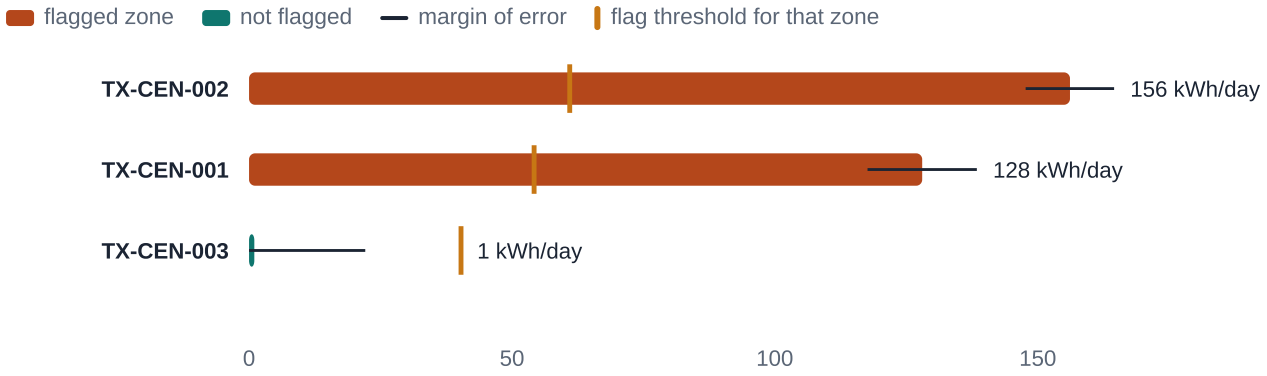


What this shows: for each day, energy the transformers supplied minus what customers' meters recorded, minus the estimate for monthly-read meters and the expected technical loss in cables and transformers. **How to read it:** a line inside the shaded band could be measurement error; a line well above it means energy is leaving the network unrecorded.

Takeaway: About 285 kWh a day goes unrecorded, well above the margin of error (±25 kWh) and steady compared with the week before (−2.5%). A steady, flat leak points to a continuous load, not a one-off.

3 · Where is it going missing?

Unexplained energy per zone this week (kWh/day)



How to read it: a zone is flagged when its unexplained energy stays above its threshold mark plus the margin of error for 7 days in a row. Thresholds differ by zone because they scale with the energy each transformer supplies (5% of supply, or 15 kWh a day, whichever is larger).

Zone	District	kWh/day	% of supply	Margin	Threshold	Status	Leak started	Daily pattern	Priority	Open cases
TX-CEN-002	Petaling	156	13%	±8	61	Flagged	2026-08-31	flat 24/7 (CV 0.05)	Normal	2
TX-CEN-001	Petaling	128	12%	±10	54	Flagged	since data start	flat 24/7 (CV 0.16)	Normal	3
TX-CEN-003	Klang	1	0.1%	±21	40	Not flagged	–	–	–	0

Takeaway: TX-CEN-002 is the priority: the largest flagged leak, flat around the clock. TX-CEN-003's 1 kWh a day sits inside its margin of error, so it is not flagged.

Zones without a transformer meter (no balance available)

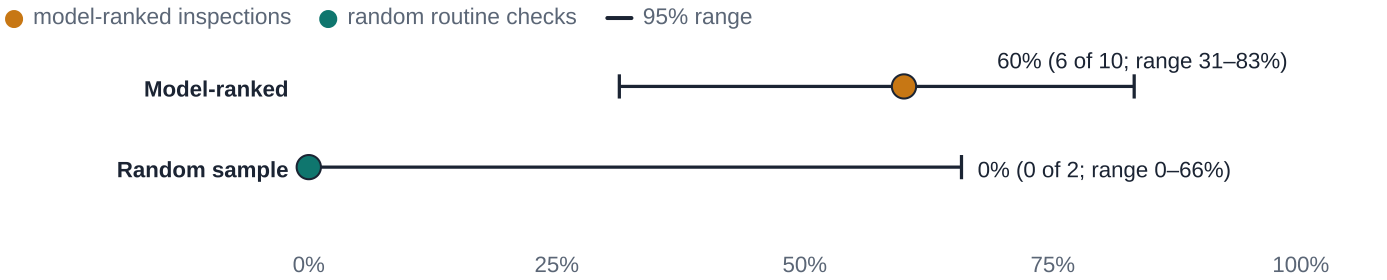
TX-CEN-004: no transformer meter, so no energy balance can be drawn; their premises are ranked on usage patterns only, at low confidence. Candidates for a transformer meter.

Value at risk (estimate)

285 kWh a day × 30 days ≈ **8,554 kWh a month**. In RM: [TNB tariff per kWh: to be supplied]. This counts all unexplained energy, including the margin of error, so it is an upper estimate; counting only energy above each zone's threshold gives about **5,067 kWh a month**.

4 · Are inspections finding theft?

Hit rate so far, with 95% ranges (% of inspections)



Inspections	Labelled	Confirmed theft	Hit rate	95% range	What it means
Model-ranked	10	6	60%	31–83%	Homes chosen by VoltGuard's ranking
Random sample	2	0	0%	0–66%	Homes picked at random in flagged zones: the fair baseline

How to read it: the hit rate is the share of inspected homes where theft was confirmed. Random routine checks show what you would find without the model, so the gap between the two rows is the model's value. The 95% range shows how much the rate could move by chance with so few inspections.

Takeaway: Model-ranked inspections look better (60% vs 0%), but the ranges overlap: 2 random check(s) are too few to prove a difference. Outcomes so far come from simulated cases, so they show how the measurement works, not real performance.

Inspection outcomes

Outcome	This week	All time
Confirmed theft	2	6
False alarm	0	5
Inconclusive	0	1
No access (not used for learning)	0	0

Cases by status now

Status	Cases	Meaning
New	2	Waiting for the supervisor
Assigned	1	Team sent
Visited	0	Outcome recorded
Reporting	2	Theft confirmed, report in approval
Closed	10	Finished

5 · Can these figures be trusted?

Data quality (meter week)

Meter-days with ≥ 90% of readings	100%
Meters read monthly (estimated daily)	15%
Zones without a transformer meter	1
Duplicate or out-of-order readings fixed	not recorded for this data source
Fallbacks in use	no transformer meter in 1 zone(s): listed without a balance (AS-2); voltage is simulated (AS-3)

Model in use

Scoring version	r0-additive/pattern-v1
Last retrained	not retrained yet; 12 labels available
Data or drift alerts this week	0
Thresholds	5% of supply or 15 kWh/day, 7 days; large load above 15%

Assumptions and limits

- **Simulated data.** This prototype runs on a simulated region built from public household profiles; voltages come from a network model, because open data hold energy readings only.
- **Technical loss is estimated** by an engineering model of cables and transformers; its error is part of the margin of error.
- **Monthly-read meters** (15% of meters) are spread across the month using similar homes' daily patterns; this widens the margin.
- **Small leaks can hide:** a tap of about 1% of a zone's supply sits inside the margin and will not flag the zone.
- **Scores rank risk;** they are not findings of theft. People decide after inspection.
- **Tariff not yet supplied,** so value at risk is shown in kWh only.
- **Zone-level only:** this report holds no premise IDs, names or addresses, and covers region CEN only.

6 · Recommended next steps

1. Manager: add random routine checks: 2 so far is too few to measure the model's real hit rate (aim for at least 10).
2. Data team: confirm the TNB tariff so value at risk can be shown in RM.

Steps are generated from fixed rules on the figures above, not written by the language model.

Glossary

Unexplained energy	Supplied by the transformer minus metered use, minus estimates for monthly-read meters and expected technical loss.
Margin of error (u)	How far unexplained energy can stray by chance, from meter accuracy, monthly-read estimates and the loss model.
Flagged zone	Unexplained energy above the threshold plus the margin of error for 7 days in a row.
Flat 24/7	The leak hardly changes across the day (coefficient of variation below 0.2), typical of mining rigs.
Hit rate	Share of inspected homes where theft was confirmed.
Random routine check	An inspection of a home picked at random in a flagged zone, used as the fair baseline.

VoltGuard AI · R0 prototype · Team F-35 · region CEN · report VG-WK-CEN-2026-W44 · data: simulated · scoring: r0-additive/pattern-v1 · scores rank risk; people decide after inspection.